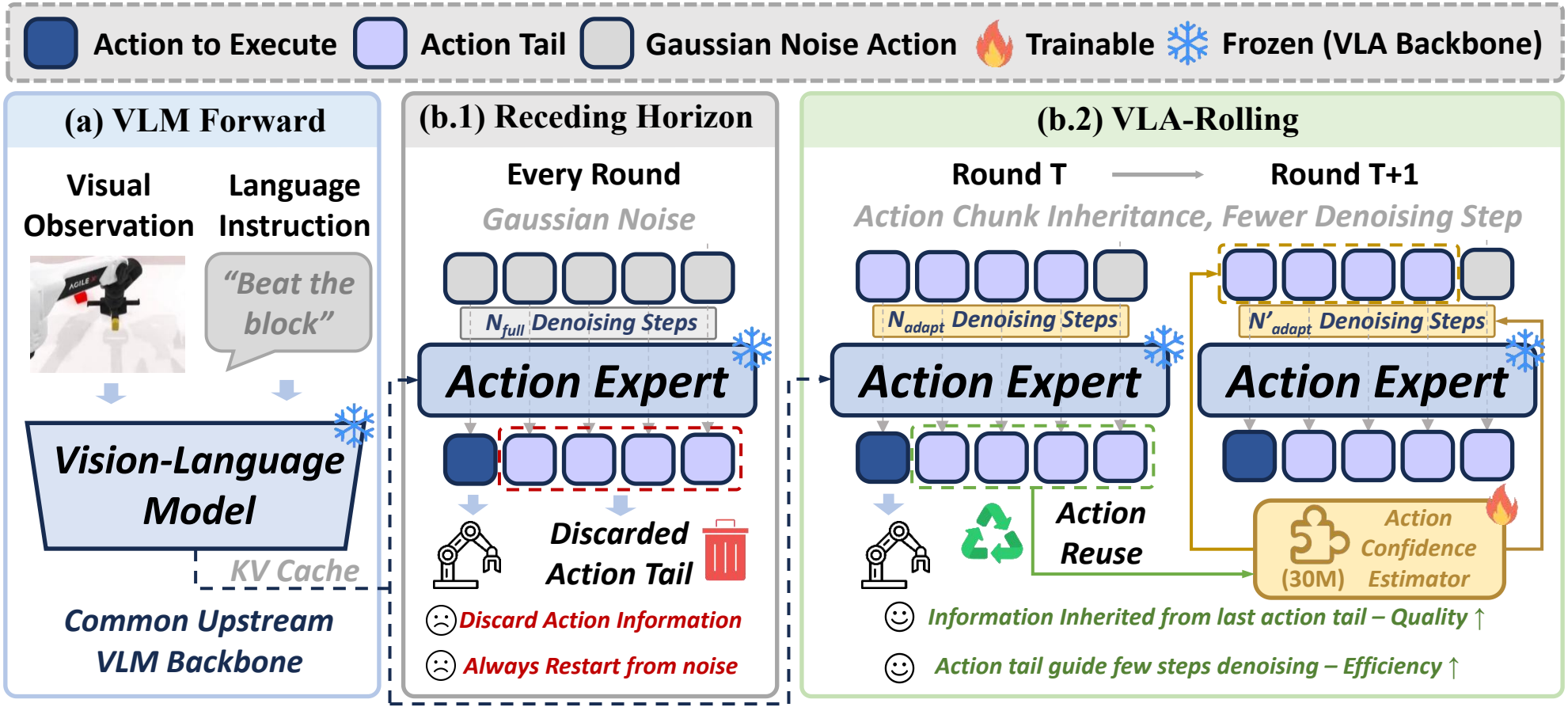

Action to Execute
Action Tail
Gaussian Noise Action
Trainable
Frozen (VLA Backbone)
(a) VLM Forward
Visual Observation
Language Instruction
"Beat the block"
Vision-Language Model
KV Cache
Common Upstream VLM Backbone
(b.1) Receding Horizon
Every Round
Gaussian Noise
N_{full} Denoising Steps
Action Expert
Discarded Action Tail
Discard Action Information
Always Restart from noise
(b.2) VLA-Rolling
Round T
Round T+1
Action Chunk Inheritance, Fewer Denoising Step
N_{adapt} Denoising Steps
N'_{adapt} Denoising Steps
Action Expert
Action Expert
Action Reuse
Action Confidence Estimator
(30M)
Information Inherited from last action tail – Quality ↑
Action tail guide few steps denoising – Efficiency ↑